

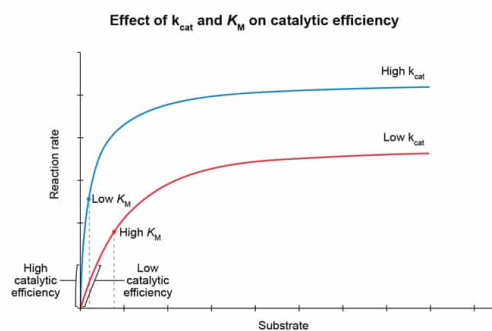
Scientists could confirm that an allosteric effector increases the catalytic efficiency of an enzyme if it has what effect on  $k_{\text{cat}}$  and  $K_{\text{m}}$ ?

- ☐ A.  $k_{\text{cat}}$  decreases,  $K_{\text{m}}$  increases [3%]
- ☐ B.  $k_{\text{cat}}$  decreases,  $K_{\text{m}}$  remains unchanged [8%]
- ✓ ☒ C.  $k_{\text{cat}}$  increases,  $K_{\text{m}}$  decreases [83%]
- ☐ D.  $k_{\text{cat}}$  remains unchanged,  $K_{\text{m}}$  increases [4%]

Correct

82% answered correctly

### Explanation:



**Catalytic efficiency** is a measure of how well an enzyme facilitates reactions at **low substrate concentrations**. When enzyme concentration is held constant, catalytic efficiency is proportional to the **initial slope** of a Michaelis-Menten curve (**when  $[S] \ll K_{\text{m}}$** ). This slope is governed by both the catalytic turnover  $k_{\text{cat}}$  and the Michaelis constant  $K_{\text{m}}$ .

Mathematically, catalytic efficiency is calculated as the **ratio of  $k_{\text{cat}}$  to  $K_{\text{m}}$** . An increase in  $k_{\text{cat}}$  increases the catalytic efficiency, and a decrease in  $K_{\text{m}}$  also increases the efficiency. Therefore, an allosteric activator that both **increases  $k_{\text{cat}}$**  and **decreases  $K_{\text{m}}$**  will increase the catalytic efficiency of an enzyme.

Conceptually, a large (steep) slope indicates an efficient enzyme because a small amount of substrate yields a high reaction rate. **Catalytic turnover ( $k_{\text{cat}}$ )** measures the number of substrate molecules converted to product per enzyme per second when the enzyme is saturated; therefore, a high  $k_{\text{cat}}$  corresponds to a high maximum rate ( $V_{\text{max}}$ ) and, by extension, a larger slope at lower substrate concentrations. The **Michaelis constant ( $K_{\text{m}}$ )** is an indicator of binding affinity. A small  $K_{\text{m}}$  indicates high affinity, allowing the reaction to approach  $V_{\text{max}}$  more rapidly (higher slope) with smaller amounts of substrate. Therefore, a small  $K_{\text{m}}$  contributes to high catalytic efficiency.

**(Choice A)** A decrease in  $k_{\text{cat}}$  and an increase in  $K_{\text{m}}$  would *decrease* catalytic efficiency.

**(Choice B)** If  $K_{\text{m}}$  is unchanged, a decrease in  $k_{\text{cat}}$  will cause a decrease in catalytic efficiency. A decrease in  $k_{\text{cat}}$  must be overcome by a proportionally greater decrease in  $K_{\text{m}}$  for catalytic efficiency to increase.

**(Choice D)** If  $k_{\text{cat}}$  is unchanged, an increase in  $K_{\text{m}}$  will decrease catalytic efficiency. An increase in  $K_{\text{m}}$  must be overcome by a proportionally greater increase in  $k_{\text{cat}}$  for catalytic efficiency to increase.

**Educational objective:**

Catalytic efficiency is a measure of an enzyme's ability to catalyze reactions at low substrate concentrations. Efficiency is calculated as  $k_{\text{cat}}/K_{\text{m}}$ , and its value can be increased by increasing  $k_{\text{cat}}$  or by decreasing  $K_{\text{m}}$ .

**Time spent:** 0 sec

**Subject:**  
Biochemistry

**QID:** 401895

**System:**  
Enzymes